

SHETH LUJ AND SIR MV COLLEGE
Cyber & Information Security

Practical No: 5

Aim: Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CLI Method:

Code:

```
Practical No 5 Cyber & Information Security T097.py - C:/Users/Omith/OneDrive/Documents/Cyber & Information Security/Practical No 5 Cyber & Information Security T097.py (3.13.1)
File Edit Format Run Options Window Help
def diffie_hellman_cli():
    print("=== Diffie-Hellman Key Exchange ===")
    print("Step 1: Setup Public Network Parameters (Interceptable by attackers)")

    try:
        p = int(input("Enter a Prime Number (p) [e.g., 23]: "))
        g = int(input("Enter a Primitive Root modulo p (g) [e.g., 5]: "))

        print("\nStep 2: Generate Private Keys (Kept secret)")
        a = int(input("Alice, enter your private key (a): "))
        b = int(input("Bob, enter your private key (b): "))

        print("\nStep 3: Calculate Public Keys (Sent over the network)")
        # Alice calculates A = (g^a) mod p
        A = pow(g, a, p)
        # Bob calculates B = (g^b) mod p
        B = pow(g, b, p)

        print("\n[*] Alice's Public Key (A) = {g}^a mod {p} = {A}")
        print("\n[*] Bob's Public Key (B) = {g}^b mod {p} = {B}")

        print("\nStep 4: Swap Public Keys & Calculate Shared Secret")
        # Alice calculates Secret = (B^a) mod p
        secret_alice = pow(B, a, p)
        # Bob calculates Secret = (A^b) mod p
        secret_bob = pow(A, b, p)

        print("\n[*] Alice calculates Shared Secret = {B}^a mod {p} = {secret_alice}")
        print("\n[*] Bob calculates Shared Secret = {A}^b mod {p} = {secret_bob}")

        if secret_alice == secret_bob:
            print("\n[✓] SUCCESS! Both parties safely agreed on the shared secret key:", secret_alice)
        else:
            print("\n[✗] ERROR! The shared secrets do not match.")

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    print("\nStep 2: Generate Private Keys (Kept secret)")
    a = int(input("Alice, enter your private key (a): "))
    b = int(input("Bob, enter your private key (b): "))

    print("\nStep 3: Calculate Public Keys (Sent over the network)")
    # Alice calculates A = (g^a) mod p
    A = pow(g, a, p)
    # Bob calculates B = (g^b) mod p
    B = pow(g, b, p)

    print("\n[*] Alice's Public Key (A) = {g}^a mod {p} = {A}")
    print("\n[*] Bob's Public Key (B) = {g}^b mod {p} = {B}")

    print("\nStep 4: Swap Public Keys & Calculate Shared Secret")
    # Alice calculates Secret = (B^a) mod p
    secret_alice = pow(B, a, p)
    # Bob calculates Secret = (A^b) mod p
    secret_bob = pow(A, b, p)

    print("\n[*] Alice calculates Shared Secret = {B}^a mod {p} = {secret_alice}")
    print("\n[*] Bob calculates Shared Secret = {A}^b mod {p} = {secret_bob}")

    if secret_alice == secret_bob:
        print("\n[✓] SUCCESS! Both parties safely agreed on the shared secret key:", secret_alice)
    else:
        print("\n[✗] ERROR! The shared secrets do not match.")

    except ValueError:
        print("\n[!] Please enter valid integer numbers.")

if __name__ == '__main__':
    diffie_hellman_cli()
```

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Output:

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
= RESTART: C:/Users/Omith/OneDrive/Documents/Cyber & Information Security/Practical No 5 Cyber & Information Security T097.py
=== Diffie-Hellman Key Exchange ===
Step 1: Setup Public Network Parameters (Interceptable by attackers)
Enter a Prime Number (p) [e.g., 23]: 2
Enter a Primitive Root modulo p (g) [e.g., 5]: 5

Step 2: Generate Private Keys (Kept secret)
Alice, enter your private key (a): 22
Bob, enter your private key (b): 23

Step 3: Calculate Public Keys (Sent over the network)
[*] Alice's Public Key (A) =  $5^{22} \bmod 23 = 1$ 
[*] Bob's Public Key (B) =  $5^{23} \bmod 23 = 1$ 

Step 4: Swap Public Keys & Calculate Shared Secret
[*] Alice calculates Shared Secret =  $1^{22} \bmod 23 = 1$ 
[*] Bob calculates Shared Secret =  $1^{23} \bmod 23 = 1$ 

[✓] SUCCESS! Both parties safely agreed on the shared secret key: 1
>>>
```

GUI Method:

Code:

```
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""GUI Method""
import flet as ft

def main(page: ft.Page):
    # --- Window Settings ---
    page.title = "Diffie-Hellman Key Exchange Studio"
    page.window.width = 750
    page.window.height = 800
    page.theme_mode = ft.ThemeMode.DARK
    page.bgcolor = "#0D1117"
    page.padding = 30
    page.scroll = ft.ScrollMode.AUTO
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER

    # Palette
    CYAN = "#00FFA3"
    PURPLE = "#A371F7"
    YELLOW = "#E3B341"
    GREEN = "#3FB950"
    CARD_BG = "#161B22"

    # --- UI Components ---
    header = ft.Row(
        controls=[
            ft.Icon(ft.Icons.SYNC_ALT, color=YELLOW, size=32),
            ft.Text("Diffie-Hellman Studio", size=28, weight=ft.FontWeight.W_800, color=ft.Colors.WHITE)
        ],
        alignment=ft.MainAxisAlignment.CENTER
    )

    # 1. Public Network Parameters Card
    p_input = ft.TextField(label="Prime Number (p)", value="23", width=200, border_color="#30363D", focused_border_color=YELLOW)
    g_input = ft.TextField(label="Base / Primitive Root (g)", value="5", width=200, border_color="#30363D", focused_border_color=YELLOW)
```

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```
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if secret_alice == secret_bob:
    log.append("[SUCCESS] Keys match! Shared Secret: {secret_alice}")
    show_toast("✅ Secure Exchange Successful!", GREEN)
else:
    log.append("[ERROR] Shared secrets do not match! Mathematical error.")
    show_toast("❌ Exchange Failed", ft.Colors.RED_700)

terminal.value = "\n".join(log)
page.update()

except ValueError:
    show_toast("⚠️ Please enter valid integers in all fields.", ft.Colors.RED_700)

# Action Button
btn_style = ft.ButtonStyle(shape=ft.RoundedRectangleBorder(radius=8), padding=20, bgcolor="#238636", color=ft.Colors.WHITE)
run_btn = ft.Button("EXECUTE KEY EXCHANGE", icon=ft.Icons.BOLT, on_click=run_exchange, style=btn_style, width=float("inf"))

# Add to page
page.add(
    header,
    ft.Divider(height=10, color="transparent"),
    network_card,
    ft.Divider(height=5, color="transparent"),
    keys_row,
    ft.Divider(height=10, color="transparent"),
    run_btn,
    ft.Divider(height=10, color="transparent"),
    terminal
)

if __name__ == "__main__":
    ft.run(main)
```

Output:

